

Po-Sheng Cheng

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Portfolio: <https://bencer3283.github.io/art/>

Master of Industrial Design, Rhode Island School of Design, Jun. 2026
Cross-enrolled at School of Engineering, Brown University
B.Sc. Bio-Mechatronics Engineering, National Taiwan University, Jan 2023

Skills

- Electrical: PCB design (Altium), STM32, Raspberry Pi, system interface (I2C, SPI, UART)
- Coding: C/C++, JavaScript (React, Express, MongoDB), Dart (Flutter), Python (PyTorch, NumPy), Git, Linux

Experiences

Design Technologist Intern, Smart Design, Jun. - Aug. 2025

Brooklyn, NY

- Developed embedded firmware (STM32) for prototyping the HMI of a household product, while working with UX design to test and fine-tune user interaction flow and hardware user interface.
- Developed testing units for motor and pneumatic mechanisms in the product, including the electrical hardware, embedded firmware and host software. (PyQt)

Product Development Apprentice, Google, Jun. - Sep. 2024

Taipei, Taiwan

- Led a team of 5 to design a Hardware-as-a-Service platform for managing lost items in public facilities. Laid out the system architecture of the hardware kiosk, user-facing app and web services.
- My prototypes of Raspberry Pi and ESP32 hardware, a React-based website, a mobile app and a RESTful API service drove the development direction from a single hardware product to an integrated platform.

Electro-mechanical Engineering Intern, Logitech, Feb. - Jun. 2022

Hsinchu, Taiwan

- Created a new breed of keyboard technology with an Arduino microcontroller-based prototype to control actuators like stepper motor and electromagnet. Also lots of CAD (Creo) and SLA/FDM prototyping on the mechanical design.
- The prototype contributed to the product development direction of the gaming keyboard team across ID, PM and EE/ME and became part of Logitech's innovation arsenal.

Projects

Haptic Belt

- Designed a custom rigid-flex PCB for a belt that uses vibration feedback to provide navigation for visually impaired people.
- Performed signal/power integrity testing on driver, interfacing ICs and vibration motors. Implemented STM32 firmware for the vibration motor driver.

MITM Router

- Built a WIFI router that brings physical, tactile interactions to the user's internet usage with the Linux networking stack on a Raspberry Pi.
- Extensive prototyping and user testing of the physical user interface was done using several sensors/actuators on a RP2040 microcontroller.

Publications

TeleShift: Teleexisting TUI for Physical Collaboration & Interaction

- Utilized ESP32 to implement a shape-transforming device with a 3D tangible user interface that enabled this work to be awarded **Best Demo** at 2022 ACM Ubicomp conference.
- Designed the robust interactive mechanisms (SolidWorks). My design for manufacturability (DFM) improvements also slashed assembling time by 80% during demo.